# "Four Bright Coins Shining at Me"

# Financial Education in Childhood, Financial Confidence in Adulthood[♦]

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**Abstract**

We analyze the relation between receiving an allowance (pocket money) in childhood and financial confidence in adulthood. We measure this confidence using self-reported financial knowledge. Our empirical exercise is based on information provided by a Dutch survey carried out in 2015. We compute our estimates by controlling for parental attitudes and by using a "within-family" fixed effect model. The results are robust and suggest a long-lasting effect of pocket money as an easily implementable and informal educational vehicle to help children acquire basic financial concepts and develop good habits, such as budgeting.

**Keywords**: pocket money; financial education; financial confidence.

**JEL**: D91; I22; J13

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[♦] We are grateful to the Editor Sharon Tennyson, two anonymous referees, as well as the participants to the 2017 Netspar International Pension Workshop and to the 2017 Cherry Blossom Financial Education Institute for their helpful comments and suggestions. We also wish to thank the EU MOPACT Grant n. 320333 for funding. In this paper use is made of data of the DNB Household Survey.

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/joca.12207

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1. **Introduction**

The role of financial literacy in enhancing people's capabilities to effectively manage their income and wealth in a life cycle perspective has long been recognized, at both theoretical and empirical level. Policy recommendations have identified in financial education programs a major instrument to help people save for their retirement, avoid taking up "too much" debt, make imprudent mortgage decisions or suffer the negative consequences of myopic financial decisions. Although the effectiveness of these programs is yet to be firmly established, the general principle that financial education can be seen as a necessary tool – though certainly not a sufficient one - to create a less unequal playing field in the economic sphere is well recognized. Despite this, researchers have found low levels of financial literacy, even among young adults (Lusardi et al. 2010).

There are, of course, various ways to help people acquire basic financial elements (*financial literacy)* and possibly to build further knowledge on these elements: from formal/compulsory programs in schools, to informal/voluntary courses specifically addressed to segments of the population considered to be "more at risk" of poor financial decisions. Parents can also contribute to improve their offspring's financial knowledge by providing children with some basic financial concepts and the opportunity to learn, as well as by involving them in youth financial literacy programs (Van Campenhout 2015).

How do children build up their attitudes towards money and saving? Are parents responsible for transferring good savings habits by nudging children towards helpful financial habits? Mandell (2008) highlights the pivotal role of parents as a key source of financial information for students at high school, finding a strong and monotonic relationship between financial literacy scores and parental education. For low and middle income households, the link between parental habits and

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saving attitudes is even more important (Grinstein-Weiss et al. 2011): Americans who had been exposed to high levels of money-management teaching in childhood by their parents show higher credit scores and lower credit card debts as adults.

Interestingly, parental socialization – that is, having parents who taught how to budget and encouraged saving – seems to be as important as financial knowledge in determining financial planning (Jorgensen and Savla 2010). Serido et al. (2010) also find that explicit parent-child socialization helps to develop sound financial behavior more than parental social status. Grohmann et al. (2015) provide a general framework to link childhood experiences within family, school and work to financial literacy, numeracy, and financial behavior. In line with the previous studies, they also find that financial socialization by parents is positively related to financial literacy.

In this paper, we look at one possible way to help children to acquire a basic familiarity with decisions involving money, i.e. the practice of "pocket money". More specifically, we test whether the children's habit of managing a little money - received more or less regularly by parents/grandparents - is related to the achievement of a greater ability to cope with financial matters later on in life. The basic idea is that the habit of properly managing some pocket money could generate a familiarity with "good" lifelong financial practices, such as budgeting.

Using Dutch data, we provide sound evidence of a positive relation between receiving an allowance during childhood and the level of (self-assessed) financial knowledge as adult: individuals who are used to receive an allowance are also more knowledgeable in adulthood. We also provide evidence that receiving an allowance in childhood is associated with a better understanding of inflation.

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This field of research is quite novel, as the previous literature has concentrated on the effects of (well-structured) financial education programs for children/students and pointed out that these courses produce a persistent impact (Bernheim et al. 2001; McCormick 2009; Loke et al. 2015). Batty et al. (2015) for the US, Romagnoli and Trifilidis (2015) for Italy, Alan and Ertac (2016) for Turkey provide evidence of long-lasting effects of basic courses taught to pupils in elementary schools. Furthermore, Gross et al. (2005) discuss the implementation and benefits of a financial literacy course at the university level. Carlin and Robinson (2012) provide evidence of higher savings, faster debt repayment and less reliance on credit for students who attended a financial literacy training. Finally, Becchetti et al. (2013) show the effects in terms of higher propensity to get economic news in the media, while Lührmann et al. (2015) find less impulsiveness in purchasing.

Very little has been written, instead, on the effect of allowances and pocket money during childhood on subsequent financial behavior. A few authors have looked at the relationship between allowances and saving behavior (Mortimer et al. 1994; Otto 2013; Webley and Nyhus 2013; Bucciol and Veronesi 2014; Brown and Taylor 2016). These studies estimate the impact of pocket money on the propensity to save to be zero (Webley and Nyhus 2013) or even negative (Brown and Taylor 2016). Nevertheless, Bucciol and Veronesi (2014) emphasize that the effectiveness of allowances may depend on its combination with other parental intervention, such as household chores.

More generally, Furnham (1999) and Furnham (2001) study parental attitudes and children behavior concerning allowances, highlighting that most parents favor clear rules around pocket money and they do not provide extra money if their children spend all their allowances. In addition to this, Holford (2016) studies the relation between pocket money and teenagers' labor

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supply in a non-cooperative game framework: he finds that parents reduce their financial support as their children increase their working hours. Furthermore, it is worth mentioning that it has been established that children are able to use sophisticated saving strategies (Otto et al. 2006).

The topic is worth further studying more in depth not only because the practice of pocket money can be a good supplement to formal financial literacy courses in school, but also because it can shed some light on the role of (good) habit formation (like acquiring some ability to plan) on saving behavior. Indeed, developing good financial habits early in life deeply affects saving choices later in adulthood (Otto and Webley 2016). Giving an allowance during childhood is likely to stimulate the ability to better manage wealth in the future: knowing in advance the total amount of money available for consumption over a given period of time forces children to plan ahead and adjust expenditure and savings accordingly. Receiving money when needed without a regular timing (for example to go to the movie theater) does not help to develop a planning consciousness. On the other hand, with a fixed amount of money, a child has to decide her priorities and plan how much to spend on each good or service without the risk of running out of money for her favorite activities. Somehow, this might be the best way to tech the concept of opportunity cost at an early age.

In this context, our paper is also related to the literature on habit persistence in saving behavior over the lifetime and across generations. These ideas can be traced back to Becker (1993) and have been investigated more recently by, among the others, Webley and Nyhus (2006), Friedline et al. (2013), Caballé and Moro-Egido (2014). Intergenerational linkages have been shown to be powerful in many circumstances. For instance, Cronqvist and Siegel (2015) use twin data to document that genetic differences explain one-third of the variation in the propensity to save across individuals. Similarly, Cesarini et al. (2010) find a strong relation between genetic

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variations and risk preferences. Ghent and Kudlyak (2015) also report intergenerational trends in household credit, while Lusardi et al. (2010) find that youth financial literacy is associated with family financial sophistication. Since it is perhaps less likely that children in lower income families receive a pocket money allowance, these children lose an opportunity to learn about budgeting and to build good financial practices. Consequently, financial education programs in primary schools may help break these intergenerational trends and increase mobility.

In other words, if high-income families are more likely to help their children to acquire basic knowledge and to form good habits, this could enhance inequality. Lusardi et al. (2016) find that 30-40% of wealth inequality can be attributed to financial knowledge. It is important, thus, that (high-quality) formal education programs are institutionalized: they may create a more leveled playing field and compensate, at least partially, for the differences that can be passed on by the families. In fact, Ghent and Kudlyak (2015) argue that economic education weakens the intergenerational linkages in the credit market (even if they do not find a similar effect for stricter financial literacy requirements in schools).

Last but not least, this analysis takes inspiration from the literature summarized in Heckman et al. (2006) and Cunha et al. (2010) on cognitive and non-cognitive abilities, as well as on the positive effects of early childhood education. This is particularly important for disadvantaged children (Heckman and Masterov 2007). Our paper contributes to this literature by stressing the value of developing financial skills early in life and by estimating their long-run returns.

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## 2. Data and descriptive statistics

### 2.1 The DHS Household Survey

The data for this analysis are drawn from the DHS Household Survey 2015, a longitudinal survey carried out every year since 1993 by CentERdata at Tilburg University and sponsored by the Dutch Central Bank. This survey was formerly known as the CentER Saving Survey. The aim of the survey is to collect information about the economic and psychological determinants of saving behaviors at the individual and household levels.

The data set is quite rich, providing detailed information about individual characteristics, employment, pensions, living conditions, mortgages, income, assets, loans, health, economic and psychological concepts. The latter part of the survey contains most of the variables relevant for our analysis: individuals' attitudes towards saving; their perceptions of their own financial situation compared to that of their peers; their own risk perception and risk aversion; their expectations about their future economic conditions; and their financial planning behaviors. A few questions are also asked about individuals' childhood and adolescence.

A characteristic of the DHS survey is that data are collected using an online questionnaire. The respondents fill in the questionnaires at their home computers, thus the questionnaires are self-administered, the respondents can answer the question at their own convenience and without the intervention of an interviewer. Households without a computer or access to the Internet are provided with a basic computer connected to the Internet. This computer is specifically designed for older people and individuals with low computer skills. CentERdata also provides technical

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assistance. The response rate at the individual level is usually high, above 70%. Participants receive a monetary compensation for filling in the questionnaire.[4]

Between April and October 2015, 2,128 households were interviewed. This random sample is representative of the Dutch population. All household members aged 16 or over were invited to complete the questionnaire, although some sections focused on certain categories only, such as the household head.

### 2.2 Descriptive statistics on allowance and financial knowledge

The data contain information on whether the interviewed person received an allowance or pocket money during childhood and on how she judges her own current financial knowledge. Data on financial knowledge have been derived from the following question:

*How knowledgeable do you consider yourself with respect to financial matters?*

1. *Not knowledgeable*
2. *More or less knowledgeable*
3. *Knowledgeable*
4. *Very knowledgeable*

The majority of the respondents in the sample do not consider themselves financially knowledgeable: around 17% of them report not to be knowledgeable, while 56% rate themselves as more or less knowledgeable. 23% consider themselves knowledgeable, while less than 4% of the respondents report very high levels of financial knowledge (Table A2 in the Online Appendix). There is a clear –statistically significant - gender gap: only 13% of men consider

[4] Teppa and Vis (2012) discuss the advantage and disadvantages of self-administered surveys and provide additional information on the dataset. CentERdata (2015) also provides additional technical details regarding the survey.

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themselves not knowledgeable with respect to financial matters, while the same figure for women is 22%.

The key question regarding childhood experiences is the following:

> *When you were between 8 and 12 years of age, did you receive an allowance from your parents then? By allowance we mean a fixed amount received on a regular basis.*
>
> 1. *Yes*
> 2. *Yes, but it was sometimes forgotten*
> 3. *Occasionally*
> 4. *No*

Among those who answered this question, around 47% received such an allowance, 7% received it but it was sometimes forgotten, 13% received it only occasionally, and 33% never received it. Female respondents were slightly less likely not to receive an allowance or to receive it only occasionally than male respondents (46% vs. 45%), although such difference is not statistically significant.

The connection between these two variables clearly indicates that self-reported financial knowledge is higher among those who received an allowance as children (Figure 1). Indeed, among the respondents who did not receive pocket money when children (or received it only occasionally), only 22.7% deemed themselves knowledgeable or very knowledgeable, while the same figure increases to 30.1% among those who (almost) always received such allowance, a statistically significant difference (t-value: -4.3). In the next sections, we show that this positive relationship between receiving an allowance in childhood and financial knowledge in adulthood is confirmed by several econometric analyses.

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**[INSERT FIGURE 1 HERE]**

## 3 Econometric Framework

### 3.1 Ordered Probit Model

Our aim is to test whether receiving, on a regular basis, an allowance between the age of 8 and 12 is positively related to financial confidence, measured as self-reported financial knowledge, later in life. As already mentioned, respondents were asked to measure how knowledgeable they consider themselves with respect to financial matters using a scale ranging from 1 to 4. Given the logical ordering of this dependent variable ($y_i$), we can estimate the following ordered probit model.

$$y_i^* = \beta_1 allowance_i + \beta_2 tbudget_i + x_i' \beta_3 + \varepsilon_i$$

$$y_j = j \; if \; \gamma_{j-1} < y_i^* < \gamma_j$$

Where $y_i^*$ is the latent variable, i.e. the underlying continuous measure of individual financial confidence, $y_i$ is the observe discrete level of such financial confidence ($y_i = 1, 2, 3, 4$), $\gamma_0 = -\infty$, $\gamma_1 = 0$, and $\gamma_4 = +\infty$. The error term $\varepsilon_i$ is assumed to have a standard normal distribution.

As far as the key independent regressor is concerned ($allowance_i$), we construct an indicator variable equal to one if the individual reported regularly receiving pocket money as a child, even if parents sometimes forgot to comply. We assign zero when the respondent reported receiving no allowance or receiving it only occasionally. Around 54% of the individuals in the relevant sample reported regularly receiving such an allowance (Table A6 in the Online Appendix).

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We have included several socio-demographic controls ($x_i'$) in the regression: gender, age, education, working and marital status, household composition, income, as well as regional indicators (a detailed description of these controls, as well as their summary statistics, is included in the Online Appendix).

More importantly, we control for parental attitudes and family background by adding an indicator variable equal to one if the respondent's (grand)parents taught her, when she was between ages 12 and 16, how to manage a little budget ($tbudget_i$). This variable should thus (partially) capture the cultural environment in which the person grew up. We later also control for additional family background characteristics, as discuss in Section 4.2.

### 3.2 Within-couple Fixed Effects

Since our relevant regressor is time-invariant, we cannot exploit the panel dimension of DHS by estimating an individual fixed effects (FE) model. However, the survey questions on economic and psychological concepts are asked to more than just one individual per household. Therefore, in addition to showing the results from the standard order probit model discussed above, we can also focus on the household head and the spouse (or the permanent unmarried partner) and use the variation within the household by adding a FE to capture all common factors between these two individuals. More specifically, we use a household fixed effects estimator and verify whether different levels of financial knowledge within the couple are due to different financial education during childhood. The idea behind this approach is that since there is assortative matching in the marriage market (Verbakel and Kalmijn 2014), partners share several individual characteristics which may affect financial knowledge. The FE model allows us to control for these unobservable components.

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Formally, we observe the same outcome (financial confidence, $y_{ht}$) twice in each household: for the household head (H) and the spouse (S). Therefore, we can write the following equation for each of the household members *t* in each interviewed household $h$.

$$y_{ht} = \beta_1 allowance_{ht} + \beta_2 tbudget_{ht} + x'_{ht}\beta_3 + \mu_h + \varepsilon_{ht} \qquad \forall t \epsilon \{H, S\}$$

Contrary to the standard FE model, here the subscript $t$ indicates different individuals in a household rather than different time periods. By taking the difference - within the same household - between the household head and the spouse, we are able to difference out $\mu_h$, i.e. to control for all the observable and unobservable factors shared by the two respondents.

As just mentioned, these include not only current household characteristics - such as household income, location, and assets – but also everything else that partners may have in common: education, behavioral traits, occupational choices. Most importantly, if partners had similar childhood experiences and were raised in similar households, we would be able to take that into account. In other words, in this case the estimate of $\beta_1$ is obtained by comparing - within households - the financial confidence of household heads and spouses, but only for those households in which only one member used to receive an allowance as a child.

## 4. Empirical Results

This section reports the estimated coefficients and marginal effects of the econometric models discussed in Section 3. We have included in the manuscript only the main results, while all additional robustness checks are available in the Online Appendix. When not reported, tables are available upon request.

### 4.1 Ordered Probit

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The estimated coefficients from the ordered probit model are reported in the first column of Table 1, while the subsequent columns contain the marginal effects on financial knowledge for the four confidence levels.

**[INSERT TABLE 1 HERE]**

Our main result is that if an individual used to receive a regular allowance as a child, she is more confident on financial issues in adulthood. In particular, this regressor reduces the probability that an individual considers herself "not knowledgeable" (Level 1) or "more or less knowledgeable" (Level 2) by 1-3 percentage points, while it increases the probability that such individual answers "knowledgeable" (Level 3) or "very knowledgeable" (Level 4) by around 1-3 percentage points.

Among the other regressors, it is interesting to note that female respondents are less likely to report high levels of financial knowledge. We have also tried to add an interaction term between allowance and gender: the coefficient is significant at a 10-percent level and negative (Table A8 in the Online Appendix). Furthermore, parenting seems to play an important role during adolescence, too. Indeed, individuals tend to have higher levels of financial knowledge if their parents or grandparents directly taught them some money management techniques. The order of magnitude is also rather large, comparable to the one of tertiary education. This suggests that being trained in budgeting during adolescence may have a substantial impact on subsequent adult behaviour, particularly if compared with other factors such as general education.

### 4.2 Sensitivity analysis

For the sake of completeness, we have also estimated an ordered logit model, finding results that are qualitatively very similar to the order probit model (Table A9 Columns 1-5 in the Online

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Appendix). It is then possible to use the test developed by Brant (1990) in order to verify the assumption that the effects of allowance on each pair of adjacent ordered outcome groups is the same, i.e. there is no heterogeneity in the relation between allowance and the log-odd ratios. It is not possible to reject this null hypothesis of "parallel regression" assumption (p-value=0.765), thus increasing our confidence in the results from the standard order logit model.[5] In addition to this, we have estimated a linear model (Table A9 Column 6). The OLS coefficient of allowance is 0.08 and it is significant, thus supporting the conclusions from the nonlinear models.

Robustness checks as the inclusion of additional controls have also been performed: adding homeownership does not substantially change the impact of receiving an allowance (Table A10). Similarly, the results do not change if we control for the number of household members instead of the number of children in the household (Table A11), or if we distinguish between employees and self-employed (Table A12).

Further, when we include as regressor age as a quadratic function, its coefficient is not statistically significant, ruling out a nonlinear impact of age (Table A13). Macroeconomic policies have also changed across generations. Individuals at different age may recall past events more or less accurately. Therefore, we have also estimated the same model as in Table 1 for two different groups in order to investigate potential differences across generations. Nevertheless, the coefficients, as well as the marginal effects, are similar when looking at individuals between age 35 to 55 (Table A14) or those older than 55 (Table A15).

[5] This test has been conducted in Stata using the code provided by Long and Freese (2014). Following Williams (2006), we have also estimated a partial proportional odd model, i.e. a generalized ordered logit model, in which the parallel-lines constraint is relaxed for the variable where it is not justified by the Brant's test (Stata code: gologit2 with option autofit(0.1)). Even using these sticter assumptions, the coefficients of allowance are positive (around 0.212) and statistically significant (p-value 0.031). Code available upon request.

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The DHS data contain additional information about the respondent's childhood and adolescence. We have already referred to one variable – i.e. whether respondent's parents taught her how to manage a small budget – that we have used to control for family background characteristics. As an additional robustness check, we have added all the other variables as regressors in our model. In particular, we have included four new indicator variables. The first one shows whether the respondent used to do small household chores in the relevant age group, i.e. between 8 and 12 years. The second measures the child's financial autonomy, i.e. whether the respondent's parents had little or no control, or on the contrary full or near full control, over the spending of the pocket money. The third one indicates whether the respondent had one or more "small" jobs when she was a teenager (12-16 years old). The fourth one indicates whether the respondent's parents or grandparents directly stimulated her to save money as a teenager. Our conclusions do not change: receiving an allowance during childhood has a statistically significant impact on the respondent's financial knowledge (Table A16). The marginal effects are similar in magnitude to those presented in Table 1. Moreover, as expected, given the multicollinearity among these variables, all these additional controls have coefficients that are not statistically different from zero.

In a symmetric way, it is interesting to note that if we remove "Parents taught budgeting" from the set of regressors, the coefficient of the pocket money becomes larger in magnitude (0.142 instead of 0.118 in Table 1). Nevertheless, the allowance's marginal effects are similar to those in Table 1 and highly statistically significant (Table A17). If we instead add an interaction term to capture potential complementarities between having received an allowance and having parents who taught how to budget, the coefficient of such term is not statistically different from zero (Table A18).

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So far we have used an indicator variable to signal whether or not an individual received a regular allowance as a child. This choice has been made not only for the sake of simplicity and in order to have a parsimonious model, given the relatively small sample size, but also because of the potential measurement error due to possible misremembering: after many years, individuals may recall whether they used to receive such an allowance; however, they may not remember very well how often this was the case. Moreover, the word "sometimes" in the option "yes, but it was sometimes forgotten" could induce different thresholds among respondents (Peracchi and Rossetti 2012). Therefore, an indicator variable attenuates these issues.

Nevertheless, we have also investigated potential heterogeneity effects, as shown in Table 2. Indeed, the model is the same as in Table 1, but we use three indicator variables for allowance instead of just one. The first variable is equal to one if the respondent always received an allowance as a child, zero otherwise; the second is one only if the respondent used to receive such an allowance "almost always", i.e. if sometimes parents forgot about it. The third variable is set to one if the respondent's parents occasionally gave her an allowance, while the groups of people who did not receive an allowance during childhood is our comparison group. Given this set up, it is possible to conclude that the group of people who always received an allowance drove the results in the previous section. Indeed, such regressors are associated with statistically significant marginal effects in Table 2. On the other hand, the other two groups are not distinguishable from those who did not get any pocket money. We can thus suggest that receiving an allowance in childhood can potentially increase financial confidence in adulthood only if parents are consistent and follow their commitment throughout the years.

**[INSERT TABLE 2 HERE]**

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### 4.3 Within-couple Fixed Effects

The estimated coefficient from a FE linear probability model is reported in Table 3[6]. Having received pocket money increases the probability of reporting some knowledge in financial matters by more than 10 percentage points. Indeed, in this model we have used as dependent variable an indicator equal to one if the respondent reported some positive level of knowledge on financial matters, zero otherwise. This effect is statistically significant and similar to the impact of allowance on the latent variable in the order probit model.

**[INSERT TABLE 3 HERE]**

It is reassuring that the estimated results from the conditional FE logit model are also qualitatively similar (Table A20 Column 1), confirming that the above result is not driven by the selected functional form. As in the previous section, we have also extended our controls by adding all the available family characteristics related to childhood and adolescence. The coefficient of allowance remains positive and highly statistically significant (Table A20 Column 2).

Furthermore, we have estimated a linear FE model with the 4-level categorical variable for financial knowledge. As shown in the second column of Table 2, the coefficient of allowance is qualitatively similar to our previous estimates, thus supporting the above conclusions.

[6] By construction, we have used in this specification a sample of individuals who are either the household head or the spouse. For comparison, we have also tried to estimate a simple order probit as in the previous paragraphs by using the same sample as the FE model and by adding an indicator equal to one if the individual is the household head (Table A19). Results do not change substantially.

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**4.4 Internal validity**

We have shown that the relation between receiving an allowance and self-reported financial knowledge is positive and significant in a large variety of econometric estimates. There are a few factors that would lead us to conclude that these results are highly suggestive that the relation between these two variables is indeed causal.

First, the pocket money we refer to was received during childhood, so it is unlikely that it is correlated with some of the covariates that affect financial literacy in adulthood. Government measures and macroeconomic shocks, for instance, may influence financial knowledge; nonetheless, these factors are not correlated with whether or not the respondent received pocket money as a child. Therefore, omitting such variables would not lead to biased estimates.

Second, the order probit model includes a large set of controls. In particular, we do control for several factors regarding the family of origin, as well as relevant experiences during childhood and adolescence.

Third, and most importantly, the within-couple fixed effect controls for all factors that are constant between partners. As already mentioned in Section 3.2, these factors could include cognitive and non-cognitive abilities, neighbors and peer effects, as well as childhood experiences. In particular, in case of positive assortative matching, we can account for the fact that partners may have grown up in similar households, thus taking into account shared factors which may be related with both the probability of receiving an allowance as a child and self-reported financial knowledge in adulthood.

However, there are a few limitations that we are not able to fully address, thus preventing us from claiming with certainty that the relation between allowance and financial knowledge is

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causal. First, we cannot rule out that, in addition to the factors included as controls such as whether the parents taught the respondent how to budget, there may be additional unobservable characteristics of the family of origin – such as personality traits - which may differ across partners and affect both the probability of receiving an allowance and financial confidence.

Second, people may recall childhood experiences differently. For example, someone with a traumatic childhood might block out certain periods (e.g., a time when they were ill, or a time when a parent lost a job and the family had to move) and would not be able to recall receiving an allowance when that happened (or vice versa). As previously discussed, we have combined whether the respondent recalled receiving an allowance always (or almost always), thus reducing this concern. Moreover, we still find an effect when focusing on younger respondents, who thus have a more recent memory of their childhood (Table A14). However, we cannot rule out the possibility that respondents may have completely forgotten about such childhood experience, and that such measurement error may be related with self-reported financial knowledge in adulthood.

As a final robustness check, we have tested the stability of our estimated coefficient of allowance in case of omitted variables by implementing the approach suggested by Altonji et al. (2005) and Oster (2017). This procedure has also been implemented with a binary outcome variable by Scott-Clayton and Minaya (2016). The key assumption is that the bias due to unobservable components is correlated with the observable controls. If we take our specification with the largest set of controls, i.e. the one in Column 2 of Table A20 including both fixed effects and all the variables regarding the respondent's family of origin, such assumption is not unrealistic, since we expect most of the bias to arise from unobserved parental characteristics. Oster's method implies that the unobservables would need to be 1.7 times as important as the

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observables to produce a zero treatment effect of allowance. Since the heuristic threshold for such ratio is 1, this result attenuates our concerns regarding potential omitted variable bias.

**4.5 External validity**

There are two concerns regarding the external validity of this study: whether the results from this analysis might be valid over time, and whether they might be the same across countries. The relation between financial confidence and pocket money might vary over time. However, as already mentioned, the coefficient associate with allowance does not change substantially when examining only individuals between age 35 and 55 rather than all respondents: the magnitude goes from 0.118 in Table 1 (Column 1) to 0.170 in Table A14 (Column 1), implying marginal effects of 1-3 percentage points versus 2-4 percentage points respectively (Columns 2-5). Similar conclusions are obtained when comparing the coefficient of allowance estimated using the whole sample (Table 1) or only respondents aged 55 or more (Table A15). Therefore, given the low heterogeneity across generations, it is likely that the relation between allowance and financial confidence is stable over time.

The Netherlands are similar in term of GDP per capita, public expenditure composition, age structure, unemployment rate and education system to several countries in Western Europe. Consequently, it is plausible that the positive relation between allowance and financial knowledge could be found in other neighboring countries. Indeed, as discussed in the introduction, researchers have found similar patterns in the Netherlands (Bucciol and Veronesi 2014) and the UK (Brown and Taylor 2016) when analyzing the impact of allowance on saving behaviors. More research is needed to verify whether the same relations hold in different developing and developed countries.

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### 4.6 Effect of allowance on inflation knowledge

In 2015, the following question on inflation was asked to a subset of the DHS sample[7]:

> *Suppose the interest rate on your savings account is 1% per year and inflation is equal to 2% per year. Could you then after 1 year buy more, exactly the same or less than today with the money in the account?*
>
> 1. *More than today*
> 2. *Exactly the same as today*
> 3. *Less than today*
> 4. *I do not say*
> 5. *I don't know*

Around 83% of the respondents answered the question correctly. Even more interestingly, such figure is greater than 85% among individuals who used to receive an allowance, while it is 81% among the others. As in the previous sections, there is a gender gap: 88% of men answered this question correctly, while only 77% of women provided the right answer. In line with the literature, women are also more likely to answer "I don't know" (11% vs 4.2%). All these differences are statistically significant.

There is a strong relation between reported and actual financial knowledge. Almost 24% of individuals who reported not to be knowledgeable with respect to financial matters answered incorrectly the above question on inflation. On the other hand, 87% of the individuals who

[7] We would like to thank Rob Alessie for sharing this survey with us. The data also contain a simple question on the effect of compound interest rates. However, more than 90% of the respondents provided the correct answer. Therefore, there is not enough variation in the sample to detect any significant effect of receiving an allowance on such form of financial knowledge.

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considered themselves knowledgeable answered it correctly. The percentage increases up to 94% among respondents who considered themselves very knowledgeable.

Despite the fact that the sample size is substantially reduced, we have estimated a Linear Probability Model and a Probit model for inflation using the same regressors as in Table 1. As reported in Table 4, both the OLS coefficient and the marginal effect of allowance in the Probit model are positive and highly statistically significant[8]. Receiving an allowance during childhood is positively associated with understanding in adulthood the effects of inflation. Indeed, people who received an allowance are 5% more likely to answer the above question correctly.

**[INSERT TABLE 4 HERE]**

Therefore, this last section provides suggestive evidence that the practice of pocket money can not only increase financial confidence, but also actual financial knowledge later in life.

## 5. Conclusions and directions for future research

This study aims at contributing to the literature on financial literacy and financial education by looking at a specific way a child can get a basic financial education and start forming good habits in financial matters, such as dealing with budget constraints and some financial planning. In line with previous findings on the importance of parental influence on financial behavior (Tang et al. 2015), we provide robust evidence of a positive relation between (regularly) receiving pocket money during childhood on the level of (self-assessed) financial knowledge in adulthood. Individuals who received a (somewhat regular) allowance in their early youth are also more

[8] We have also tried to estimate a within-couple fixed-effect model as in Table 3. However, the coefficient of allowance is undistinguishable from zero due to the small sample size: the conditional logit model uses only 94 observations. Similarly, we have also tried to add an interaction term between allowance and female in the OLS and Probit model from Table 4, but its coefficient is not statistically significant.

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likely to become financially knowledgeable adults. This conclusion is also supported by the relation between receiving such an allowance and knowledge about inflation in adulthood. These findings offer evidence of the importance of providing basic financial education to children in order not only to give them essential financial knowledge and help them acquire some planning capabilities, but also to develop some good permanent financial habits.

Our result is consistent with previous work on financial literacy suggesting that: a) more financially literate households are less vulnerable to under-saving and therefore are better equipped for retirement (van Rooij et al. 2011) and b) its uneven distribution within the population is a fact that explains a significant part of wealth inequality (Lusardi et al. 2017). Giving pocket money to offspring is a potentially simple and inexpensive way to provide children with basic financial literacy and prepare them for future financial choices. While this runs counter to the arguments against financial education on the basis of its high costs and limited benefits (Willis 2011; Fernandes et al. 2014), it also indirectly advocates for school programs delivered to all children as a way to smooth the differences among families and create a more leveled playing field.

Further research is needed to investigate, possibly with cross-country comparisons, whether receiving an allowance during childhood affects educational achievements - specifically math knowledge and abilities - as well as financial decisions later in life. Gender bias in the pocket money practice and its implications are also worth investigating.

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**Tables and Figures**

**Figure 1**

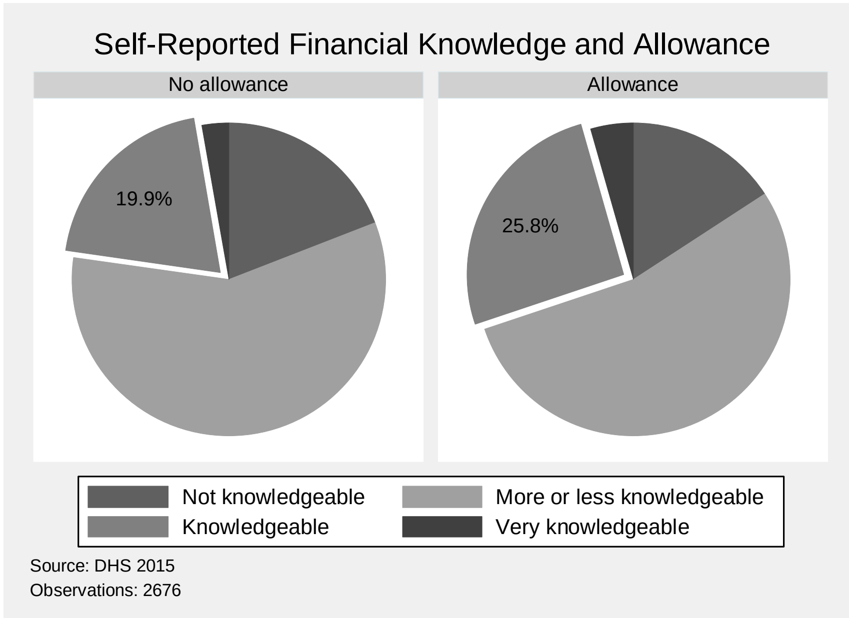


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**Table 1: Ordered probit.**

| | (1) Coeff | (2) Level 1 | (3) Level 2 | (4) Level 3 | (5) Level 4 |
|---|---|---|---|---|---|
| Financial knowledge | | | | | |
| Allowance | 0.118$^{**}$ | -0.028$^{**}$ | -0.011$^{**}$ | 0.030$^{**}$ | 0.009$^{**}$ |
| | (0.055) | (0.013) | (0.005) | (0.014) | (0.004) |
| Female | -0.321$^{***}$ | 0.076$^{***}$ | 0.029$^{***}$ | -0.080$^{***}$ | -0.025$^{***}$ |
| | (0.052) | (0.012) | (0.006) | (0.013) | (0.005) |
| Age | -0.003 | 0.001 | 0.000 | -0.001 | -0.000 |
| | (0.002) | (0.001) | (0.000) | (0.001) | (0.000) |
| Tertiary education | 0.216$^{***}$ | -0.051$^{***}$ | -0.020$^{***}$ | 0.054$^{***}$ | 0.017$^{***}$ |
| | (0.054) | (0.013) | (0.005) | (0.013) | (0.005) |
| Log(Individual Gross Income) | 0.039$^{***}$ | -0.009$^{***}$ | -0.004$^{***}$ | 0.010$^{***}$ | 0.003$^{***}$ |
| | (0.011) | (0.002) | (0.001) | (0.003) | (0.001) |
| Working | -0.027 | 0.006 | 0.002 | -0.007 | -0.002 |
| | (0.065) | (0.015) | (0.006) | (0.016) | (0.005) |
| Parents taught budgeting | 0.196$^{***}$ | -0.046$^{***}$ | -0.018$^{***}$ | 0.049$^{***}$ | 0.015$^{***}$ |
| | (0.066) | (0.016) | (0.006) | (0.016) | (0.005) |
| Married | 0.147$^{**}$ | -0.035$^{**}$ | -0.013$^{**}$ | 0.037$^{**}$ | 0.012$^{**}$ |
| | (0.059) | (0.014) | (0.006) | (0.015) | (0.005) |
| Number of children in the HH | -0.008 | 0.002 | 0.001 | -0.002 | -0.001 |
| | (0.028) | (0.007) | (0.003) | (0.007) | (0.002) |
| Threshold 1 | -0.466$^{**}$ | | | | |
| | (0.197) | | | | |
| Threshold 2 | 1.136$^{***}$ | | | | |
| | (0.197) | | | | |
| Threshold 3 | 2.406$^{***}$ | | | | |
| | (0.203) | | | | |
| Regional dummies | Yes | Yes | Yes | Yes | Yes |
| Observations | 2,014 | 2,014 | 2,014 | 2,014 | 2,014 |

This table analyzes the relation between receiving an allowance during childhood and self-reported financial knowledge in adulthood. Standard errors clustered at the household level in parentheses. The first column shows the estimated coefficients from the order probit. The reported marginal effects are divided into four columns. Level 1 refers to the probability of reporting "Not Knowledgeable". Level 2 refers to the probability of reporting "More or less knowledgeable". Level 3 refers to the probability of reporting "Knowledgeable". Level 4 refers to the probability of reporting "Very Knowledgeable". Source: DHS 2015.

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

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**Table 2: Ordered probit. Heterogeneity allowance.**

| | (1) Coeff | (2) Level 1 | (3) Level 2 | (4) Level 3 | (5) Level 4 |
|---|---|---|---|---|---|
| Financial knowledge | | | | | |
| Allowance - Always | 0.153** | -0.036** | -0.014** | 0.038** | 0.012** |
| | (0.062) | (0.015) | (0.006) | (0.015) | (0.005) |
| Allowance - Almost always | 0.078 | -0.019 | -0.007 | 0.019 | 0.006 |
| | (0.113) | (0.027) | (0.010) | (0.028) | (0.009) |
| Allowance - Occasionally | 0.077 | -0.018 | -0.007 | 0.019 | 0.006 |
| | (0.082) | (0.019) | (0.008) | (0.020) | (0.006) |
| Female | -0.319*** | 0.076*** | 0.029*** | -0.080*** | -0.025*** |
| | (0.052) | (0.012) | (0.006) | (0.013) | (0.005) |
| Age | -0.002 | 0.001 | 0.000 | -0.001 | -0.000 |
| | (0.002) | (0.001) | (0.000) | (0.001) | (0.000) |
| Tertiary education | 0.214*** | -0.051*** | -0.020*** | 0.053*** | 0.017*** |
| | (0.054) | (0.013) | (0.005) | (0.013) | (0.005) |
| Log(Individual Gross Income) | 0.039*** | -0.009*** | -0.004*** | 0.010*** | 0.003*** |
| | (0.011) | (0.002) | (0.001) | (0.003) | (0.001) |
| Working | -0.029 | 0.007 | 0.003 | -0.007 | -0.002 |
| | (0.065) | (0.015) | (0.006) | (0.016) | (0.005) |
| Parents taught budgeting | 0.190*** | -0.045*** | -0.017*** | 0.047*** | 0.015*** |
| | (0.066) | (0.016) | (0.006) | (0.016) | (0.005) |
| Married | 0.150** | -0.035** | -0.014** | 0.037** | 0.012** |
| | (0.059) | (0.014) | (0.006) | (0.015) | (0.005) |
| Number of children in the HH | -0.006 | 0.002 | 0.001 | -0.002 | -0.000 |
| | (0.028) | (0.007) | (0.003) | (0.007) | (0.002) |
| Threshold 1 | -0.433** | | | | |
| | (0.199) | | | | |
| Threshold 2 | 1.169*** | | | | |
| | (0.200) | | | | |
| Threshold 3 | 2.439*** | | | | |
| | (0.205) | | | | |
| Regional dummies | Yes | Yes | Yes | Yes | Yes |
| Observations | 2,014 | 2,014 | 2,014 | 2,014 | 2,014 |

This table analyzes the relation between receiving an allowance during childhood (one indicator variable for each possible answer) and self-reported financial knowledge in adulthood. Standard errors clustered at the household level in parentheses. The first column reports the estimated coefficients from the order probit. The reported marginal effects are divided into four columns. The Level 1 refers to the probability of reporting "Not Knowledgeable". The Level 2 refers to the probability of reporting "More or less knowledgeable". The Level 3 refers to the probability of reporting "Knowledgeable". The Level 4 refers to the probability of reporting "Very Knowledgeable". Source: DHS 2015.

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

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**Table 3: Within-household linear fixed effects model**

| | (1)<br>2 Categories | (2)<br>4 Categories |
|---|---|---|
| Financial knowledge | | |
| Allowance | 0.1024** | 0.1921** |
| | (0.0476) | (0.0816) |
| Female | -0.0835* | -0.1296 |
| | (0.0458) | (0.0918) |
| Age | -0.0070 | -0.0211 |
| | (0.0067) | (0.0145) |
| Tertiary education | 0.0429 | 0.1709* |
| | (0.0572) | (0.0988) |
| Log(Individual Gross Income) | 0.0005 | 0.0187 |
| | (0.0068) | (0.0125) |
| Working | -0.0726 | -0.1947* |
| | (0.0542) | (0.1005) |
| Parents taught budgeting | -0.0100 | -0.0232 |
| | (0.0591) | (0.0943) |
| Household head | 0.0566 | 0.1886** |
| | (0.0421) | (0.0856) |
| Constant | 1.2018*** | 3.0575*** |
| | (0.4067) | (0.8518) |
| Observations | 1,953 | 1,953 |
| Within-$R^2$ | 0.07 | 0.14 |
| Overall-$R^2$ | 0.01 | 0.03 |
| Average obs per ind | 1.23 | 1.23 |

This table analyzes the relation between receiving an allowance during childhood and self-reported financial knowledge in adulthood. In Column 1, the dependent variable is an indicator equal to one if the respondent reported some positive level of knowledge on financial matters, zero otherwise. In Column 2, the dependent variable is the same original 4-level categorical variable used in Tables 1 and 2. Standard errors clustered at the household level in parentheses. Source: DHS 2015.

$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

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**Table 4: Effect of allowance on inflation knowledge**

| | (1)<br>OLS | (2)<br>Probit |
|---|---|---|
| Inflation knowledge | | |
| Allowance | 0.0491*** | 0.0474*** |
| | (0.0182) | (0.0175) |
| Female | -0.0720*** | -0.0772*** |
| | (0.0172) | (0.0160) |
| Age | 0.0028*** | 0.0027*** |
| | (0.0008) | (0.0008) |
| Tertiary education | 0.1151*** | 0.1308*** |
| | (0.0169) | (0.0204) |
| Log(Individual Gross Income) | 0.0129*** | 0.0089*** |
| | (0.0042) | (0.0028) |
| Working | -0.0297 | -0.0280 |
| | (0.0222) | (0.0202) |
| Parents taught budgeting | 0.0506** | 0.0470** |
| | (0.0224) | (0.0201) |
| Married | 0.0268 | 0.0260 |
| | (0.0211) | (0.0194) |
| Number of children in the HH | 0.0100 | 0.0082 |
| | (0.0106) | (0.0100) |
| Constant | 0.5180*** | |
| | (0.0716) | |
| Regional dummies | Yes | Yes |
| Observations | 1,674 | 1,674 |

This table analyzes the relation between receiving an allowance during childhood and financial knowledge. The binary dependent variable indicates whether the respondent answered correctly the question on inflation. Column 1 reports the estimates from the Linear Probability Model, while the estimates from the Probit Model are shown in Column 2. Standard errors clustered at the household level in parentheses. Source: DHS 2015.
$^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

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